

Nicholas Jennings

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----- EDUCATION -----

Stanford University

PhD Computer Science

2025 - Present

- Rotating with Maneesh Agrawala, Kayvon Fatahalian, & James Landay

University of California, Berkeley

M.S. Electrical Engineering & Computer Science

2023 - 2024 (GPA: 4.0)

- Advised by Prof. Björn Hartmann

B.A. Computer Science, Applied Mathematics

2019 - 2023 (GPA: 3.94)

- High Distinction in General Scholarship

----- RESEARCH EXPERIENCE -----

Stanford University

Stanford, California

Graduate Research (PhD)

Sep 2025 - Present

- Rotating with Maneesh Agrawala, James Landay, and Kayvon Fatahalian. I'm working on developing guardrails to allow for better developer control over runtime code generation, as well as a methodology for how these guardrails might be used.

University of California, Berkeley

Berkeley, California

Graduate Research (MS)

Aug 2023 - Aug 2024

- Thesis project mentored by Prof. Björn Hartmann and James Smith. I Developed a LLM-based runtime code generation system for virtual environments, then conducted quantitative evaluations and interviews with game developers to explore how this technology could impact them. **Published at UIST'24, Best Paper Award**
- Mentored by Prof. Ren Ng, I worked on an ongoing research project to model color space as experienced by human tetrachromats (people with 4 cone types instead of the usual 3). We built a custom printer setup capable of printing Ishihara-style dot tests to detect tetrachromacy. **Results Published at SIGGRAPH'24, Honorable Mention**
- Mentored by Prof. Sarah Chasins, I worked on a research project exploring the use of randomness in computer-generated art. We built an extension to p5.js to enable new ways of creating/modifying random variables, and studied the effects of this tool on creative coders.

Undergraduate Researcher, Berkeley Institute of Design

Sep 2021- May 2023

- Mentored by Björn Hartmann and James Smith. I worked on the GeneratiVR project where I created a Virtual Reality based sorting tool for use in Generative Design workflows. I designed novel spatial interactions for filtering parametric design spaces. **Published at CHI'22 and ICED'23**
- Mentored by Prof. Björn Hartmann and James Smith, I worked on a research project concerning multi-body bimanual interactions in VR. We created novel interaction techniques for selection and locomotion using Unity3D, developed a mathematical model to describe properties of the interaction, and conducted a user study.

Undergraduate Research Assistant, BLUES lab

Sep 2021- Sep 2022

- Mentored by Eleonora Losiouk. I worked on a project studying the malware ecosystem of virtualized android apps. I built Gradle scripts for virtualizing malwares and assessing their detectability by commercial antivirus apps, and used Bash, Gradle, Docker, and Python scripts to create an semi-automated system for identifying virtualization-based malwares from virus databases, allowing for more in-depth analysis.

KTH Royal Institute of Technology

Stockholm, Sweden

Research Exchange Student

June - Aug 2024

- Exchange program with UC Berkeley

- Mentored by Prof. Andrii Matviienko, worked on a research project to explore how signifiers impact the affordances of beyond-real VR interactions. I recreated and modified popular VR interaction techniques using compute shaders, and conducted and analyzed data from a user study. The resulting paper was **published at VRST'25**

California State University, San Marcos

San Marcos, CA

Computer Science Research Intern

Jan - Mar 2019

- Aided in a qualitative study on alternate methods of computer file structure representation. Used the Unity Engine to create a three-dimensional file visualization system compatible with GitHub

University of Michigan

Ann Arbor, MI

Computer Science Research Intern

May - June 2017

- Collected probate information from local records as part of a larger project to create a national database
- Used the Puppeteer web scraping library to successfully develop and deploy web scraping software to automate probate data extraction

----- PUBLICATIONS -----

- **Nicholas Jennings**, Andrii Matviienko [Reimagining Go-Go: Effects of Signifiers in VR Selection Techniques](#) VRST'25
- (**Best Paper**) **Nicholas Jennings**, Han Wang, Isabel Li, James Smith, and Björn Hartmann. [What's the Game, then? Opportunities and Challenges for Runtime Behavior Generation](#) UIST 2024
- (**Honorable Mention**) Jessica Lee, **Nicholas Jennings**, Varun Srivastava, Ren Ng. [Theory of Human Tetrachromatic Color Experience and Printing](#) SIGGRAPH 2024
- Ananya Nandy, James Smith, **Nicholas Jennings**, Mike Kuniavsky, Bjoern Hartmann. [VR or Not? Investigating Interface Type and User Strategies for Interactive Design Space Exploration](#) ICED 2023
- **Nicholas Jennings**, Ananya Nandy, Xinyi Zhu, Yuting Wang, Fanping Sui, James Smith, Bjoern Hartmann. [GeneratiVR: Spatial Interactions in Virtual Reality to Explore Generative Design Spaces](#) CHI 2022

----- WORK EXPERIENCE -----

Sandia National Laboratories

Albuquerque, New Mexico

Computer Scientist

Sep 2024 - June 2025

- Conducted research related to user interfaces for satellites

Amazon (AWS)

Seattle, WA

SWE Intern

May - Aug 2023

- Designed and implemented a gamification extension to an internal iOS application, with a native AWS REST service as a backend using services such as Lambda, DynamoDB, and SNS.
- Created an Augmented Reality app component using RealityKit

SWE Intern

May - Aug 2022

- Designed and implemented a service which used AWS services to send scheduled notifications to specified subsets of mobile app customers.
- Designed the service for easy integration with future tools, and documented how these extensions should be implemented.

----- OTHER PROJECTS -----

I Need to Get Stronger (C#, Unity)

[Link](#)

- A platform fighting game where monologuing about your powers gives your character those powers!

Firework Simulation (HLSL, C#, Unity)

[Link](#)

- Custom firework/smoke simulation and renderer

ASCII graphics renderer (C++)

[Github Link](#)

- First person camera / controller using ASCII characters, runnable in terminal

TutoriVR (C#, Unity)

[Link](#)

- Implemented a tutorial extension for VR applications for use with the OpenBrush VR painting tool

----- ADDITIONAL SKILLS -----

Languages: Java, C#, C++, Python, Swift, HLSL, HTML/CSS, TypeScript, SQL

Technologies: Unity3D, AWS, DynamoDB, JUnit, Docker, Git, Kanban, Blender, RealityKit